

Distributed state estimation with event-triggered measurement sampling^{☆,☆☆}

Irene Perez-Salesa^{ID*}, Rodrigo Aldana-López, Carlos Sagüés

Departamento de Informática e Ingeniería de Sistemas (DIIS) and Instituto de Investigación en Ingeniería de Aragón (I3A), Universidad de Zaragoza, María de Luna 1, Zaragoza, 50018, Spain

ARTICLE INFO

Keywords:

Distributed state estimation
Sensor networks
Kalman filtering
Asynchronous measurements
Event-triggered sampling and communication
Stochastic systems

ABSTRACT

In this work, we focus on distributed state estimation under event-triggered measurement sampling and estimator-to-estimator communication. We design a distributed Kalman-like filter, with fully asynchronous transmissions of measurements and estimates. The estimator nodes leverage the implicit information from not receiving new sensor measurements between events, resulting in stable estimates for any transmission sequence. Moreover, we show that the performance of the centralized Kalman–Bucy filter with full measurement data can be approximated arbitrarily well with our event-triggered solution, by tuning the event thresholds and the consensus gain in the filter, while reducing communication.

1. Introduction

Distributed sensing and estimation have attracted significant attention in recent years, finding applications in areas such as robot formation control, environmental monitoring and large-scale industrial processes. Thus, many works have been devoted to developing distributed implementations of popular state estimators, with Kalman filters being a prominent example [1]. In distributed filtering, several estimator nodes collectively observe the state of a plant, computing state estimates by using a local sensor measurements and information received from neighboring nodes. Relevant advances have been achieved, such as obtaining distributed formulations that converge to the solution of the centralized counterpart, e.g., [2–4], and developing event-triggered filters to reduce communication among estimator nodes [5–7].

However, several issues remain open. One important limitation is the assumption of synchronicity in the distributed filter. In most discrete-time formulations, it is implicitly assumed that sampling and filtering at all nodes follow the same synchronized discretization period. Even when event-triggered communication is considered (see [1,8] and the references therein), the underlying updates are typically still based on a common global clock. Such an assumption may be unrealistic in practical applications. Most works that consider asynchronous measurement sampling resort to a “data synchronization” strategy to still apply a synchronous filter [9–11]. An exception can be found in [12], which is limited to systems without noise, and requires the asynchronous period to be upper-bounded.

In addition, event-triggered communication is often applied only to the estimator-to-estimator channel, while continuous availability of local measurements is still assumed. For cases where sensors and estimator nodes are not co-located, this assumption

[☆] This article is part of a Special issue entitled: ‘TC 1.5 Networked systems (IFAC WC 2026)’ published in Nonlinear Analysis: Hybrid Systems.

^{☆☆} This work was supported via projects PID2021-124137OB-I00 and PID2024-159279OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by ERDF/EU, via project REMAIN S1/1.1/E0111 (Interreg Sudoe Programme, ERDF), by the Gobierno de Aragón, Spain under Project DGA T45-23R, and by Spanish grant PPU20/03134. Grant reference BG24/00121 funded by MICIU/AEI/10.13039/501100011033.

* Corresponding author.

E-mail addresses: i.perez@unizar.es (I. Perez-Salesa), raldana@unizar.es (R. Aldana-López), csagues@unizar.es (C. Sagüés).

may be impractical, and event-triggered communication could also reduce the number of transmissions in this channel. However, few works have considered this setup. In [13–15], the event-triggered condition is checked periodically, matching the step of a synchronous filter. Moreover, some restrictive assumptions are made on the noise and observability of the system. Finally, H_∞ filters are designed in [16,17] under event-triggered measurement transmissions. These works assume that sensors exchange information with each other, having enough capabilities to aggregate the measurements from neighbors and transmit them to the estimator node, and even run a local filter (in [17]), while the estimator nodes do not communicate with each other.

To the best of our knowledge, there is no prior work addressing distributed Kalman filtering under event-triggered measurement sampling. Note that extending the existing centralized event-triggered Kalman filters to the distributed case is not trivial. Many works consider triggering conditions that require the sensor to run a local filter [18–20], which assumes computational capabilities and knowledge of the dynamical model of the system at the sensor side. On the other hand, simple triggering conditions such as the send-on-delta [21] may cause a divergence of the estimation uncertainty if only the model-based prediction is used by the estimator during inter-event intervals, as exemplified in [22]. To overcome this issue, some event-triggered Kalman filters exploit the implicit information in *not* triggering an event [22–26], but most of them still adopt synchronous filter updates.

Motivated by this discussion, this work aims to address the current limitations in distributed Kalman filtering with the following contributions: (1) We develop a distributed filter with fully asynchronous transmissions, where measurement sampling and estimator-to-estimator communication do not need to be synchronized, by adopting a continuous-time filter formulation. (2) We consider that each sensor transmits information to an estimator node according to a simple event-triggering condition. The estimator leverages these measurements and the implicit information between events to provide stable estimates, even in the absence of new measurement data. (3) We show that event-triggered communication can also be adopted in the estimator-to-estimator channel in our setup while maintaining the stability of the distributed estimator. (4) We prove the stability of the filter in mean squared error, and we show that the performance of our distributed filter with reduced communication can approximate that of the centralized optimal filter with full information by tuning the event thresholds and consensus gain.

1.1. Notation

The normal distribution of mean μ and covariance Σ is denoted by $\mathcal{N}(\mu, \Sigma)$. The notation $\|\cdot\|$ denotes the Euclidean norm for vectors, or the induced norm for matrices, and $\|\mathbf{x}\|_{\mathbf{M}}^2 = \mathbf{x}^\top \mathbf{M} \mathbf{x}$ is the weighted norm. For a matrix $\mathbf{M}$, we write $\mathbf{M} > (\geq) \mathbf{0}$ to indicate that it is positive definite (semidefinite, respectively). The column and block-diagonal compositions of vectors or matrices with indexes $i = 1, 2, \dots, N$ is given by $\text{col}_{i=1}^N(\cdot_i)$ and $\text{diag}_{i=1}^N(\cdot_i)$. The vector of ones with dimension $n \times 1$ is given by $\mathbb{1}_n$, $\mathbf{U}_n = \mathbb{1}_n \mathbb{1}_n^\top$, and the $n \times n$ identity matrix is represented by $\mathbf{I}_n$. The symbol $\otimes$ is the Kronecker product and $\text{sign}(\cdot)$ is the sign function, with $\text{sign}(0) = 0$, $\text{sign}(x) = 1$ if $x > 0$ and $\text{sign}(x) = -1$ if $x < 0$. The operators $\mathbb{P}\{\cdot\}$ and $\mathbb{E}\{\cdot\}$ denote probability and expectation, respectively. The abbreviation *w.p.1* means “with probability 1”.

2. Problem statement

A network with N estimator and sensor nodes collaboratively observes a system

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{w}(t), \quad \forall t \geq 0, \quad (1)$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ is the state to be estimated and $\mathbf{w}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{W}(t))$ is a disturbance affecting the system. The initial state follows $\mathbf{x}(0) \sim \mathcal{N}(\mathbf{x}_0, \mathbf{P}_0)$.

The i th sensor obtains measurements $\mathbf{y}_i(t) \in \mathbb{R}^{m_i}$, which take the form

$$\mathbf{y}_i(t) = \mathbf{C}_i(t)\mathbf{x}(t) + \mathbf{v}_i(t), \quad (2)$$

with $\mathbf{v}_i(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{V}_i(t))$. The disturbances $\mathbf{w}(t)$ and $\mathbf{v}_i(t)$ at the sensors are uncorrelated. Aggregating all measurements as $\mathbf{y}(t) = \text{col}_{i=1}^N(\mathbf{y}_i(t)) \in \mathbb{R}^m$, the global measurement obtained by the sensor network can be compactly written as

$$\mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) + \mathbf{v}(t), \quad (3)$$

where $\mathbf{C}(t) = \text{col}_{i=1}^N(\mathbf{C}_i(t))$ and $\mathbf{v}(t) = \text{col}_{i=1}^N(\mathbf{v}_i(t))$ with zero mean and covariance $\mathbf{V}(t) = \text{diag}_{i=1}^N(\mathbf{V}_i(t))$.

The i th estimator node receives measurements from the i th sensor through a communication network only at a series of discrete event instants $t \in \{\tau_k^i\}_{k=0}^\infty$ according to a local event-triggering mechanism at the sensor side, where τ_k^i refers to the k th event-triggered by the i th sensor. Therefore, from the estimator's point of view, only the received measurements $\{\mathbf{y}_i(\tau_k^i)\}_{k=0}^\infty$ are known.

The estimator nodes communicate their local estimates to neighboring estimators. Their communication network is described by a connected and undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V} = \{1, 2, \dots, N\}$ being the node set representing the N estimator nodes, and the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denoting the communication channels between nodes. The notation $\mathcal{N}_i$ represents the neighbors of the i th estimator. The graph topology is described by its 0–1 adjacency matrix $\mathbf{A}_{\mathcal{G}}$ and, equivalently, by its Laplacian matrix $\mathbf{L}_{\mathcal{G}}$ with algebraic connectivity $\lambda_{\mathcal{G}}$. To lighten the communication load, each estimator node may also transmit its estimate only at a series of event instants, which we denote $t \in \{\ell_k^i\}_{k=0}^\infty$, via another event-triggering mechanism.

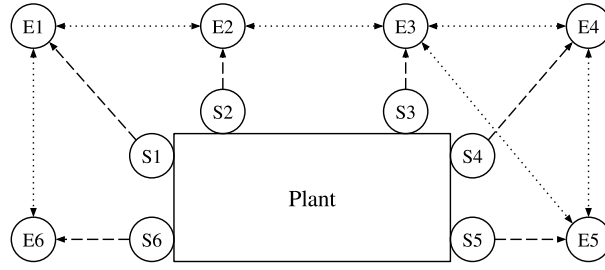


Fig. 1. Distributed state estimation setup with $N = 6$ estimation nodes (E) and sensors (S). Dashed lines represent event-triggered measurement transmissions from sensors to estimators, and dotted lines are event-triggered communication links between estimators.

Problem 1. Each estimator node i must obtain an estimate $\hat{x}_i(t)$ of the state $x(t)$, using the event-triggered measurements received from the i th sensor, as well as the information exchanged with neighboring estimator nodes $j \in \mathcal{N}_i$. Moreover, a *bounded uncertainty* for the estimates must be maintained in the absence of new sensor data for any sequence of events, despite the stochastic disturbances.

An example of the problem setup for a network with $N = 6$ estimator nodes and sensors is shown in Fig. 1. Note that the transmission sequences are independent for each sensor and estimator node, i.e., all transmissions are asynchronous.

We make the following assumptions for the system, which are standard in Kalman filtering literature to ensure the stability of the filter [27]. Note that we merely require global observability for the system using the aggregate measurement (3).

Assumption 1. The global pair $(A(t), C(t))$ is uniformly completely observable and $(A(t), W(t)^{1/2})$ is uniformly completely controllable.

Assumption 2. There exist constants $a, c \geq 0$ such that $\|A(t)\| \leq a$ and $\|C(t)\| \leq c$; $0 < w_1 \leq w_2$ fulfilling $w_1 I_n \leq W(t) \leq w_2 I_n$, and $0 < v_1 \leq v_2$ such that $v_1 I_m \leq V(t) \leq v_2 I_m$, for all $t \geq 0$.

3. Event-triggered measurement sampling

We start by introducing the event-triggering mechanism used by the sensor nodes to decide when to transmit information, as well as the implicit information that can be extracted by the estimator when the triggering condition is not met.

3.1. Event-triggering mechanism

The i th sensor decides its transmission instants via an event-triggering mechanism of the form

$$\tau_{k+1}^i = \inf \{t > \tau_k^i \mid \|y_i(t) - y_i(\tau_k^i)\| \geq \varepsilon_i(t)\}, \quad (4)$$

where $\varepsilon_i(t) > 0$ is the event threshold, possibly containing auxiliary dynamics. In order for the estimator to successfully infer implicit information from the triggering condition (4) and produce stable estimates, we consider the following requirements. First, the estimator node receiving the measurements from the i th sensor must have knowledge of $\varepsilon_i(t)$, $\forall t \geq 0$. Second, the event threshold fulfills $\varepsilon_i(t) \leq \varepsilon_{\max}$, $\forall t \geq 0$, for some $\varepsilon_{\max} > 0$.

Note that these requirements can be easily met when the event threshold is set to a constant value. In addition, the dynamic event-triggering mechanism from [22], where the dynamic threshold depends on the evolution of the transmitted measurements, also enables the extraction of implicit information despite the time-varying threshold, and fulfills the boundedness condition with probability 1, despite the event threshold being affected by the unbounded Gaussian measurement noise [22, Remark 1]. For such triggering mechanism, the stability guarantees in this work would hold with probability 1, instead of holding strictly for any noise realization.

For simplicity, we now make the following assumption:

Assumption 3. There is no Zeno behavior under the event-triggering mechanisms, i.e., a strictly positive time interval exists between two consecutive events.

We will first establish some theoretical results regarding the filter performance under this assumption, isolating its convergence and stability properties. Then, in Section 7, we will formally prove that Assumption 3 indeed holds with probability 1 despite the stochastic noise present in the system, and we will detail an easily implementable time regularization strategy to enforce a minimum inter-event time, discussing its theoretical and practical impact.

3.2. Implicit information between events

Between events, the lack of measurement data can be mitigated by exploiting the implicit information in the non-fulfillment of the event-triggering condition. The event-triggering mechanism (4) can be equally expressed as

$$\tau_{k+1}^i = \inf \{t > \tau_k^i \mid \|\mathbf{y}_i(t) - \mathbf{y}_i(\tau_k^i)\|_{\mathbf{E}_i(t)}^2 \geq 1\}, \quad (5)$$

with $\mathbf{E}_i(t) = (\varepsilon_i(t)^2 \mathbf{I}_{m_i})^{-1}$. This means that the next event τ_{k+1}^i is triggered when the measurement $\mathbf{y}_i(t)$ leaves an ellipsoid $\mathcal{E}_i(t)$, given by

$$\mathcal{E}_i(t) = \{\mathbf{y}_i(t) \in \mathbb{R}^{m_i} : \|\mathbf{y}_i(t) - \mathbf{y}_i(\tau_k^i)\|_{\mathbf{E}_i(t)}^2 < 1\}, \quad (6)$$

noting that $\mathbf{y}_i(\tau_k^i)$ is the center of the ellipsoid, and the event threshold $\varepsilon_i(t)$ determines its size.

For an event-triggering mechanism of the form (5), [24, Lemma 1] states that the posterior probability density function (PDF) of the state conditioned on not triggering an event at a time $t \in (\tau_k^i, \tau_{k+1}^i)$ is the same as the posterior PDF conditioned to receiving a measurement $\mathbf{y}_i'(t) = \mathbf{y}_i(\tau_k^i)$ originated from a virtual measurement channel

$$\mathbf{y}_i'(t) = \mathbf{C}_i(t)\mathbf{x}(t) + \mathbf{v}_i(t) - \mathbf{v}_i(t), \quad (7)$$

where $\mathbf{v}_i(t) \in \mathbb{R}^{m_i}$ is a random disturbance that is independent from $\mathbf{x}(t)$ and $\mathbf{v}_i(t)$, and uniformly distributed in the ellipsoid $\mathcal{E}_i(t)$. The virtual measurement noise fulfills $\mathbb{E}\{\mathbf{v}_i(t) - \mathbf{v}_i(t)\} = \mathbf{0}$, due to $\mathbf{v}_i(t)$, $\mathbf{v}_i(t)$ being zero-mean and independent, and its noise covariance is given by

$$\mathbf{M}_i(t) = \text{cov}\{\mathbf{v}_i(t) - \mathbf{v}_i(t)\} = \mathbf{V}_i(t) + \frac{\varepsilon_i(t)^2}{m_i + 2} \mathbf{I}_{m_i}, \quad (8)$$

recalling that the covariance of a uniform random variable contained in the ellipsoid (6) is given by $((m_i + 2)\mathbf{E}_i(t))^{-1}$.

Note that the actual measurement for $t \in (\tau_k^i, \tau_{k+1}^i)$ can be written as $\mathbf{y}_i(t) = \mathbf{y}_i(\tau_k^i) + \boldsymbol{\eta}_i(t)$, where the difference $\boldsymbol{\eta}_i(t)$ is unknown but contained within the ellipsoid $\mathcal{E}_i(t)$ around $\mathbf{y}_i(\tau_k^i)$. Therefore, the virtual measurement can be seen as an approximation of the actual measurement using only the information available to the estimator, where the center of the ellipsoid, $\mathbf{y}_i(\tau_k^i)$, is taken as the measurement value between events, and the measurement uncertainty is augmented to take into account the ellipsoid's size according to the event threshold $\varepsilon_i(t)$. Moreover, the result from [24, Lemma 1] ensures that the virtual measurement model is valid to design a suitable estimator, as the case with no new measurement data between events can be treated equally to receiving the virtual measurement from a Bayesian filtering perspective.

3.3. Measurement handling

From the point of view of the estimator, the available sets of measurement information to compute state estimates are given by the actual measurements $\mathbf{y}_i(t) = \mathbf{y}_i(\tau_k^i)$ generated by (2) with noise covariance $\mathbf{V}_i(t)$ received at discrete event instants $t \in \{\tau_k^i\}_{k=0}^\infty$, and the continuous virtual measurements $\mathbf{y}_i'(t) = \mathbf{y}_i(\tau_k^i)$ originated by (7) with covariance $\mathbf{M}_i(t)$, for $t \in (\tau_k^i, \tau_{k+1}^i)$. Incorporating this information into the filter would require hybrid continuous-discrete updates as in [22].

To reduce the complexity of the design, we will consider a simplified measurement model of the form

$$\bar{\mathbf{y}}_i(t) = \mathbf{C}_i(t)\mathbf{x}(t) + \mathbf{v}_i(t) - \mathbf{v}_i(t) = \mathbf{y}_i(\tau_k^i), \quad t \in [\tau_k^i, \tau_{k+1}^i), \quad (9)$$

with noise covariance $\mathbf{M}_i(t) \forall t \geq 0$, i.e., as if the virtual measurement model was valid for all $t \geq 0$. Note that this model overestimates the noise covariance at events, since $\mathbf{V}_i(t) \leq \mathbf{M}_i(t)$, but the conservativeness from this approximation is negligible due to the sparsity of event instants.

Then, at an arbitrary time instant t , the filter computes an estimate of the state $\mathbf{x}(t)$ based on the information set

$$\bar{\mathcal{I}}_t = \bigcup_{i=1}^N \{\bar{\mathbf{y}}_i(s), \forall s \in [0, t]\}, \quad (10)$$

which represents the set of all causal measurements of the form (9) obtained up until time t .

Note that the measurement $\bar{\mathbf{y}}_i(t)$ is defined $\forall t \geq 0$ by maintaining a constant value between events, while its covariance $\mathbf{M}_i(t)$ is also defined $\forall t \geq 0$. Thus, we will adopt a continuous-time filter formulation to produce a continuous estimate $\hat{\mathbf{x}}(t)$ of the continuous state of system (1).

4. Auxiliary centralized filter

Before we proceed with the design of the distributed filter, we introduce an auxiliary centralized formulation to aid us in establishing our formal results. Define $\tau_t^i = \max\{\tau_k^i \leq t\}$ as the most recent event triggered by the i th sensor prior to time t . Let $\bar{\mathbf{y}}(t) = \text{col}_{i=1}^N(\bar{\mathbf{y}}_i(t)) = \text{col}_{i=1}^N(\mathbf{y}_i(\tau_t^i)) = \mathbf{y}(\tau_t)$ be the global measurement available to the centralized filter, defined by keeping the last transmitted value from each node constant between events (recalling the piecewise-constant measurement model in (9)), with measurement noise covariance $\mathbf{M}(t) = \text{diag}_{i=1}^N(\mathbf{M}_i(t))$. Then, the centralized filter under event-triggered measurements is given by

$$\mathbf{K}(t) = \mathbf{P}(t)\mathbf{C}(t)^\top \mathbf{M}(t)^{-1}, \quad (11a)$$

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}(t)\hat{\mathbf{x}}(t) + \mathbf{K}(t)(\mathbf{y}(\tau_t) - \mathbf{C}(t)\hat{\mathbf{x}}(t)), \quad (11b)$$

$$\dot{\mathbf{P}}(t) = \mathbf{A}(t)\mathbf{P}(t) + \mathbf{P}(t)\mathbf{A}(t)^\top + \mathbf{W}(t) - \mathbf{K}(t)\mathbf{M}(t)\mathbf{K}(t)^\top, \quad (11c)$$

where $\mathbf{K}(t)$ is the filter gain, $\hat{\mathbf{x}}(t)$ is the centralized estimate of the state $\mathbf{x}(t)$ and $\mathbf{P}(t)$ is the filter's estimated error covariance matrix. Moreover, the filter converges to an asymptotic solution for $\mathbf{P}(t)$ in the linear time-invariant (LTI) case, given by

$$\mathbf{0} = \mathbf{A}\mathbf{P}_\infty + \mathbf{P}_\infty\mathbf{A}^\top + \mathbf{W} - \mathbf{P}_\infty\mathbf{C}^\top\mathbf{M}^{-1}\mathbf{C}\mathbf{P}_\infty. \quad (12)$$

Therefore, (11) can be further simplified to the asymptotic filter:

$$\begin{aligned} \mathbf{K}_\infty &= \mathbf{P}_\infty\mathbf{C}^\top\mathbf{M}^{-1}, \\ \dot{\hat{\mathbf{x}}}(t) &= \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{K}_\infty(\mathbf{y}(\tau_t) - \mathbf{C}\hat{\mathbf{x}}(t)). \end{aligned} \quad (13)$$

We can now state the following result for the auxiliary centralized filter.

Lemma 1. *Let Assumptions 1 to 3 hold. Consider the filter (11) along with the measurement set $\bar{I}_t$ from (10) for the system (1). Then, the filter is uniformly asymptotically stable and the best linear unbiased estimator (BLUE), providing estimates $\hat{\mathbf{x}}(t) = \arg\min_{\hat{\mathbf{x}} \in \mathbb{R}^n} \text{tr}(\mathbf{P}(t))$ with $\mathbf{P}(t) = \text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}(t) \mid \bar{I}_t\}$.*

Proof. The result follows from noticing that the filter (11) is a Kalman-Bucy filter [27] over the measurement $\bar{\mathbf{y}}(t)$ with measurement noise covariance $\mathbf{M}(t)$, and therefore it inherits its properties. $\square$

Remark 1. Note that Gaussianity of the noise processes is not required for the derivation of the Kalman-Bucy filter as the minimum variance linear estimator (see the problem statement in [27]).

5. Distributed filter with event-triggered measurements

We now propose a distributed counterpart to the filter (11), extending the design from [4]¹ to incorporate the event-triggered measurement transmissions. In this section, we assume full communication between estimator nodes, but we will introduce event-triggered communication in this channel in Section 6.

Each estimator node obtains its own estimate $\hat{\mathbf{x}}_i(t)$ of the state $\mathbf{x}(t)$ and an estimated error covariance matrix $\mathbf{P}_i(t)$ according to the following:

$$\mathbf{K}_i(t) = N\mathbf{P}_i(t)\mathbf{C}_i(t)^\top\mathbf{M}_i(t)^{-1}, \quad (14a)$$

$$\dot{\hat{\mathbf{x}}}_i(t) = \mathbf{A}(t)\hat{\mathbf{x}}_i(t) + \mathbf{K}_i(t)(\mathbf{y}_i(\tau_t^i) - \mathbf{C}_i(t)\hat{\mathbf{x}}_i(t)) + \kappa\mathbf{P}_i(t) \sum_{j \in \mathcal{N}_i} (\hat{\mathbf{x}}_j(t) - \hat{\mathbf{x}}_i(t)), \quad (14b)$$

$$\dot{\mathbf{P}}_i(t) = \mathbf{A}(t)\mathbf{P}_i(t) + \mathbf{P}_i(t)\mathbf{A}(t)^\top + \mathbf{W}(t) - \mathbf{P}_i(t)\hat{\mathbf{Z}}_i(t)\mathbf{P}_i(t), \quad (14c)$$

$$\hat{\mathbf{Z}}_i(t) = N\mathbf{C}_i(t)^\top\mathbf{M}_i(t)^{-1}\mathbf{C}_i(t) - \mathbf{Q}_i(t), \quad (14d)$$

$$\dot{\mathbf{Q}}_i(t) = \alpha \sum_{j \in \mathcal{N}_i} \phi(\hat{\mathbf{Z}}_i(t) - \hat{\mathbf{Z}}_j(t), \xi, \gamma), \quad (14e)$$

with $\sum_{i=1}^N \mathbf{Q}_i(0) = \mathbf{0}$, where the function

$$\phi(\bullet, \xi, \gamma) = (|\bullet|^{1-\gamma} + |\bullet|^{1+\gamma} + \xi)\text{sign}(\bullet) \quad (15)$$

is applied element-wise to the matrix, and $\kappa, \alpha, \xi, \gamma$ are parameters of the filter to be appropriately chosen. Ideally, the filter is initialized to $\hat{\mathbf{x}}_i(0) = \mathbf{x}_0$, $\mathbf{P}_i(0) = \mathbf{P}_0$, but a suboptimal initialization can be used when the initial state is unknown, without compromising the asymptotic behavior of the filter.

In particular, (14a) is the filtering gain, which is akin to the one in the centralized filter (11a) but uses only the local matrices $\mathbf{C}_i(t)$, $\mathbf{M}_i(t)$ and includes the factor N . This gain design has been used in prior works that consider full measurement availability to design a distributed filter that recovers the estimates of the centralized counterpart [2,4]. The corresponding state estimation update is given by (14b), which incorporates a consensus term featuring the received information from neighboring estimator nodes $j \in \mathcal{N}_i$. The covariance matrix update (14c) aims to recover the covariance matrix $\mathbf{P}(t)$ from (11c) in a distributed fashion, thanks to the auxiliary fixed-time consensus protocol (14d)–(14e).² This protocol serves to obtain estimates $\hat{\mathbf{Z}}_i(t)$ of the quantity $\mathbf{C}(t)^\top\mathbf{M}(t)^{-1}\mathbf{C}(t) = \sum_{i=1}^N \mathbf{C}_i(t)^\top\mathbf{M}_i(t)^{-1}\mathbf{C}_i(t)$, which appears in the centralized formulation for $\mathbf{P}(t)$ and contains global information of the sensor model matrices and event thresholds, that may be unknown from the point of view of each node in the network.

¹ We will refer to the formal results from this work as well as [28], which extended the proof of convergence to the optimal solution for the LTV case and provided a sufficient value for the consensus gain in the filter.

² Note that (14e) has a discontinuous right-hand side. Thus, solutions $\hat{\mathbf{Z}}_i(t)$ of (14d)–(14e) must be understood in the sense of Filippov [29].

Remark 2. In the LTI case, $\mathbf{P}_i(t)$ can be directly replaced by $\mathbf{P}_\infty$ from (12), such that the gain (14a) is constant and only (14b) needs to be run online. $\mathbf{P}_\infty$ can be computed in a distributed form by running (14d)–(14e), which obtains $\dot{\mathbf{Z}}_i(t) = \mathbf{C}^\top \mathbf{M}^{-1} \mathbf{C}$ in a fixed time $T_{\max}$. Then, each node can solve (12) locally. Another approach is to use the *push-sum* algorithm proposed in [2] to pre-compute $\mathbf{P}_\infty$. Note that the number of nodes N can also be obtained from distributed protocols such as [30].

We require the following additional assumptions.

Assumption 4. Let $\mathbf{Z}_i(t) = N \mathbf{C}_i(t)^\top \mathbf{M}_i(t)^{-1} \mathbf{C}_i(t)$. There is a constant $L \geq 0$ such that $\|\dot{\mathbf{Z}}_i(t)\| \leq L$, $\forall t \geq 0$.

Assumption 5. The parameters of the distributed filter (14) are chosen such that $\alpha > 0$, $\gamma \in (0, 1)$, $\xi \geq 2L/(\alpha\sqrt{\lambda_G})$, and κ such that the matrix

$$\mathbf{A}^*(t, \kappa) = \text{diag}_{i=1}^N (\mathbf{A}(t) - \mathbf{K}_i(t) \mathbf{C}_i(t)) - \kappa \text{diag}_{i=1}^N (\mathbf{P}_i(t)) (\mathbf{L}_G \otimes \mathbf{I}_n) \quad (16)$$

yields asymptotically stable dynamics for a system $\dot{\mathbf{e}}(t) = \mathbf{A}^*(t, \kappa) \mathbf{e}(t)$.

Remark 3. Assumption 4 and the choice of parameters α , γ , ξ in Assumption 5 are needed to ensure the convergence of the fixed-time consensus protocol (14d)–(14e) [4, Lemma 2]. Moreover, by [4, Lemma 3], there exists a value κ_0 such that Assumption 5 holds for any $\kappa > \kappa_0$.

In the following, we state the main results for our distributed filter under event-triggered communication. We first introduce a lemma relating the performance of the filter to the measurement model (9) used for the design, and then we establish the stability of the filter considering the actual event-triggered measurements.

Lemma 2. Let Assumptions 1 to 5 hold. Consider the distributed filter (14) for the state of system (1), with the measurement set (10). Then, the following properties are complied:

- (1) *Unbiasedness:* $\mathbb{E}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t) \mid \bar{\mathcal{I}}_t\} = \mathbf{0}$, $\forall t \geq 0$.
- (2) *Stability:* $\text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t) \mid \bar{\mathcal{I}}_t\}$ and $\mathbf{P}_i(t)$ are uniformly bounded $\forall t \geq 0$, for any sequence of events.
- (3) *Convergence:* As $\kappa \rightarrow \infty$, $\text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t) \mid \bar{\mathcal{I}}_t\}$ and $\mathbf{P}_i(t)$ converge to $\mathbf{P}(t)$ from the BLUE centralized filter (11) asymptotically, $\forall i \in \mathcal{V}$.

Proof. The proof follows directly from noting that (11) and (14) correspond to the centralized Kalman-Bucy filter and the distributed filter from [4] for the assumed measurements $\bar{\mathbf{y}}_i(t)$ with noise covariance $\mathbf{M}_i(t)$ from (9)–(10). Then, the lemma holds according to [4, Theorems 1-2] and [28, Theorem 2]. $\square$

Remark 4. Note that $\mathbf{P}(t) = \text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}(t) \mid \bar{\mathcal{I}}_t\}$ in the centralized filter (11), but this equality does not hold in general for $\text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t) \mid \bar{\mathcal{I}}_t\}$ and $\mathbf{P}_i(t)$ in the distributed filter (14). In fact, $\mathbf{P}_i(t)$ converges to $\mathbf{P}(t)$ regardless of the choice of κ , but this choice is important for the convergence of $\text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t) \mid \bar{\mathcal{I}}_t\}$. Still, for large values of κ , $\mathbf{P}_i(t)$ resembles $\text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t) \mid \bar{\mathcal{I}}_t\}$, as formalized in item 3 of Lemma 2.

Theorem 1. Let Assumptions 1 to 5 hold. Consider the distributed filter (14) for the state of system (1), with the measurements (2) transmitted according to the event-triggering mechanism (4). The following items hold:

- (1) The estimation error $\mathbf{e}_i(t) = \mathbf{x}(t) - \hat{\mathbf{x}}_i(t)$ is uniformly bounded in mean squared error.
- (2) If $\epsilon_{\max} \rightarrow 0$, $\kappa \rightarrow \infty$, then both $\text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t)\}$ and $\mathbf{P}_i(t)$, $\forall i \in \mathcal{V}$, converge asymptotically to $\mathbf{P}(t)$ from a centralized Kalman-Bucy filter with the full global measurements (3).

Proof. We start by proving item 1. We can compute the error dynamics of the i th estimator as $\dot{\mathbf{e}}_i(t) = \dot{\mathbf{x}}(t) - \dot{\hat{\mathbf{x}}}_i(t)$, considering (1) and (14b). By design of the triggering condition (4), we have $\mathbf{y}_i(\tau_i^j) = \mathbf{y}_i(t) - \boldsymbol{\eta}_i(t)$, where $\|\boldsymbol{\eta}_i(t)\| \leq \epsilon_i(t) \leq \epsilon_{\max}$, $\forall t \geq 0$. Combining these, and with $\hat{\mathbf{x}}_j(t) - \hat{\mathbf{x}}_i(t) = \mathbf{e}_j(t) - \mathbf{e}_i(t)$, we obtain

$$\dot{\mathbf{e}}_i(t) = (\mathbf{A}(t) - \mathbf{K}_i(t) \mathbf{C}_i(t)) \mathbf{e}_i(t) + \mathbf{w}(t) - \mathbf{K}_i(t) \mathbf{v}_i(t) + \mathbf{K}_i(t) \boldsymbol{\eta}_i(t) - \kappa \mathbf{P}_i(t) \sum_{j \in \mathcal{N}_i} (\mathbf{e}_j(t) - \mathbf{e}_i(t)), \quad (17)$$

similarly to the analysis in [4,28]. Defining $\mathbf{e}(t) = \text{col}_{i=1}^N (\mathbf{e}_i(t))$ and $\boldsymbol{\eta}(t) = \text{col}_{i=1}^N (\boldsymbol{\eta}_i(t))$, we write the global error dynamics,

$$\dot{\mathbf{e}}(t) = \mathbf{A}^*(t, \kappa) \mathbf{e}(t) + \mathbb{I}_N \otimes \mathbf{w}(t) - \text{diag}_{i=1}^N (\mathbf{K}_i(t)) \mathbf{v}(t) + \text{diag}_{i=1}^N (\mathbf{K}_i(t)) \boldsymbol{\eta}(t), \quad (18)$$

where $\mathbf{A}^*(t, \kappa)$ is defined in Assumption 5, recalling that it is an asymptotically stable dynamical matrix with a suitable choice of κ , as discussed in Remark 3. Note that the dynamics (18) are affected by stochastic disturbances $\mathbf{w}(t)$, $\mathbf{v}(t)$ and by an unknown but bounded variable $\boldsymbol{\eta}(t)$. To ease the analysis, considering the two different types of disturbances that affect the error dynamics, we split the error as $\mathbf{e}(t) = \mathbf{e}^s(t) + \mathbf{e}^b(t)$, where $\mathbf{e}^s(t)$ is the error term influenced by the stochastic components and $\mathbf{e}^b(t)$ by the bounded ones. In particular, their respective dynamics are

$$\begin{aligned} \dot{\mathbf{e}}^s(t) &= \mathbf{A}^*(t, \kappa) \mathbf{e}^s(t) + \mathbb{I}_N \otimes \mathbf{w}(t) - \text{diag}_{i=1}^N (\mathbf{K}_i(t)) \mathbf{v}(t), \\ \dot{\mathbf{e}}^b(t) &= \mathbf{A}^*(t, \kappa) \mathbf{e}^b(t) + \text{diag}_{i=1}^N (\mathbf{K}_i(t)) \boldsymbol{\eta}(t). \end{aligned} \quad (19)$$

For the stochastic part, we have $\mathbf{e}^s(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{E}^s(t))$, where the zero mean is due to the initialization $\hat{\mathbf{x}}_i(t) = \mathbf{x}_0$, which gives $\mathbb{E}\{\mathbf{e}(0)\} = \mathbf{0}$, and $\mathbf{w}(t), \mathbf{v}(t)$ being zero mean Gaussian noises. The covariance $\mathbf{E}^s(t)$ is given by

$$\dot{\mathbf{E}}^s(t) = \mathbf{A}^*(t, \kappa) \mathbf{E}^s(t) + \mathbf{E}^s(t) \mathbf{A}^{*\top}(t, \kappa) + \mathbf{U}_N \otimes \mathbf{W}(t) + \text{diag}_{i=1}^N(\mathbf{K}_i(t)) \mathbf{V}(t) \text{diag}_{i=1}^N(\mathbf{K}_i(t))^\top. \quad (20)$$

Note that $\mathbf{W}(t)$ and $\mathbf{V}(t)$ are uniformly bounded according to [Assumption 2](#). Moreover, $\mathbf{K}_i(t)$ is also bounded by this assumption and recalling the boundedness of $\mathbf{P}_i(t)$ from [Lemma 2](#). From the stable dynamics and bounded matrices, it follows that $\mathbf{E}^s(t)$ has a uniform bound $\bar{\mathbf{E}}^s$, such that $\mathbf{E}^s(t) \leq \bar{\mathbf{E}}^s$, $\forall t \geq 0$. Similarly, for the error term $\mathbf{e}^b(t)$, the disturbance $\boldsymbol{\eta}(t)$ fulfills $\|\boldsymbol{\eta}(t)\| \leq \sqrt{N} \varepsilon_{\max}$, $\forall t \geq 0$. This boundedness along with the stable dynamics from $\mathbf{A}^*(t, \kappa)$ implies that a uniform bound $\bar{E}^b$, dependent on the value of $\varepsilon_{\max}$, can be found such that $\|\mathbf{e}^b(t)\| \leq \bar{E}^b$, $\forall t \geq 0$. Combining these results, it follows that

$$\mathbb{E}\{\mathbf{e}(t)^\top \mathbf{e}(t)\} = \mathbb{E}\{(\mathbf{e}^s(t) + \mathbf{e}^b(t))^\top (\mathbf{e}^s(t) + \mathbf{e}^b(t))\} = \text{tr}(\mathbf{E}^s(t)) + \|\mathbf{e}^b(t)\|^2 \leq \text{tr}(\bar{\mathbf{E}}^s) + (\bar{E}^b)^2, \quad \forall t \geq 0, \quad (21)$$

completing the proof of uniform mean-squared-error boundedness for the estimation error.

The proof of item 2 follows as a consequence of the convergence from item 3 of [Lemma 2](#), along with noting that the full measurement information case is approximated as $\varepsilon_{\max} \rightarrow 0$. $\square$

6. Multi-channel event-triggered filter

We extend our proposal to the multi-channel event-triggered communication case, by endowing each estimator node with its own event-triggering mechanism to decide when to transmit its local estimate to neighboring nodes. The goal is to obtain a distributed filter with fully asynchronous and reduced transmissions in both communication channels. To ease the presentation and analysis, in this section we consider the particular case for LTI systems:

Assumption 6. The observed system and measurement model are LTI, i.e., $\mathbf{A}, \mathbf{W}, \mathbf{C}_i, \mathbf{V}_i, \varepsilon_i$ are constant.

Note that, in this case, only the state estimates $\hat{\mathbf{x}}_i(t)$ need to be communicated among estimator nodes, and the asymptotic form of the filter with constant $\mathbf{P}_\infty$ can be used, as detailed in [Remark 2](#).

The transmission instants $\{\ell_k^i\}_{k=0}^\infty$ for the estimator nodes, which are independent of the events $\{\tau_k^i\}_{k=0}^\infty$ triggered at the sensors, are chosen as

$$\ell_{k+1}^i = \inf\{t > \ell_k^i \mid \|\hat{\mathbf{x}}_i(t) - \hat{\mathbf{x}}_i(\ell_k^i)\| \geq \delta_i(t)\}, \quad (22)$$

where $0 < \delta_i(t) \leq \delta_{\max}$, $\forall t \geq 0$, for some $\delta_{\max} > 0$.

Then, the distributed LTI filter with multi-channel event-triggered communication is given by

$$\mathbf{K}_i = N \mathbf{P}_\infty \mathbf{C}_i^\top \mathbf{M}_i^{-1}, \quad (23a)$$

$$\dot{\hat{\mathbf{x}}}_i(t) = \mathbf{A} \hat{\mathbf{x}}_i(t) + \mathbf{K}_i (\mathbf{y}_i(\tau_t^i) - \mathbf{C}_i \hat{\mathbf{x}}_i(t)) + \kappa \mathbf{P}_\infty \sum_{j \in \mathcal{N}_i} (\hat{\mathbf{x}}_j(\ell_t^j) - \hat{\mathbf{x}}_i(t)), \quad (23b)$$

where $\hat{\mathbf{x}}_j(\ell_t^j)$ with $\ell_t^j = \max\{\ell_k^j \leq t\}$ is the most recent estimate received from the neighbor j prior to time t .

Remark 5. An event-triggered implementation of an auxiliary consensus protocol in (14) to compute $\mathbf{P}_i(t)$ for linear time-varying (LTV) systems could also be achieved through similar ideas as in [31].

Theorem 2. Let [Assumptions 1](#) to [6](#) hold. Consider the distributed filter (23) for the state of system (1) with the measurements (2), under the event-triggering mechanisms (4) and (22). Then,

- (1) The estimation error $\mathbf{e}_i(t) = \mathbf{x}(t) - \hat{\mathbf{x}}_i(t)$ is uniformly bounded in mean squared error.
- (2) If $\delta_{\max} \rightarrow 0$, $\varepsilon_{\max} \rightarrow 0$, $\kappa \rightarrow \infty$, then $\mathbb{E}\{\mathbf{e}_i(t)\} \rightarrow \mathbf{0}$ and $\text{cov}\{\mathbf{x}(t) - \hat{\mathbf{x}}_i(t)\} \rightarrow \mathbf{P}_\infty$ from a centralized Kalman-Bucy filter with full measurement information (3) asymptotically.

Proof. With the introduction of the event-triggering mechanism, the error dynamics (18) are modified as follows

$$\dot{\mathbf{e}}_i(t) = (\mathbf{A} - \mathbf{K}_i \mathbf{C}_i) \mathbf{e}_i(t) + \mathbf{w}(t) - \mathbf{K}_i \mathbf{v}_i(t) + \mathbf{K}_i \boldsymbol{\eta}_i(t) - \kappa \mathbf{P}_\infty \sum_{j \in \mathcal{N}_i} (\mathbf{e}_i(t) - \mathbf{e}_j(t)) + \kappa \mathbf{P}_\infty \sum_{j \in \mathcal{N}_i} \mathbf{d}_j(t), \quad (24)$$

by noting that $\hat{\mathbf{x}}_j(\ell_t^j) = \hat{\mathbf{x}}_j(t) - \mathbf{d}_j(t)$, with $\|\mathbf{d}_j(t)\| \leq \delta_j(t) \leq \delta_{\max}$, $\forall t \geq 0$, by design of the triggering condition (22). That is, another bounded disturbance is added to the error dynamics, recalling the proof of [Theorem 1](#). By the same arguments, it is evident that the estimation error remains bounded in mean squared error.

For item 2, as $\delta_{\max} \rightarrow 0$, the perturbation $\mathbf{d}_j(t)$ vanishes, with the error dynamics remaining as in the case analyzed in [Theorem 1](#), from which the result follows. $\square$

The results from [Theorems 1–2](#) indicate that stable state estimates can be obtained by our distributed filter under event-triggered communication in the sensor and estimator channels, while reducing the communication load. Moreover, by appropriately tuning the event thresholds and consensus gain, the estimation performance of each estimator in the distributed setup with reduced communication can be made arbitrarily close to that of a centralized Kalman-Bucy filter with full measurement information.

7. Strategies for Zeno-freeness

So far, we have provided our theoretical results while assuming the absence of Zeno behavior under the event-triggering mechanisms (4)–(22). In this section, we formally prove that [Assumption 3](#) holds with probability 1 for our event-triggered distributed filters, under some mild considerations. In addition, since only a probabilistic guarantee can be given under the stochastic noise present in the system, we also discuss an alternative time regularization strategy that can be incorporated to the event-triggering mechanism to strictly enforce a minimum inter-event time. Finally, we comment on its effect on the overall performance of the setup.

7.1. Probabilistic avoidance of Zeno behavior

This section formally establishes conditions ensuring the absence of Zeno behavior. Since the mechanism in (4) is directly influenced by stochastic measurement noise, whereas the mechanism in (22) acts on filtered estimates that incorporate neighbor information, each triggering rule is analyzed separately.

Theorem 3. *Let [Assumptions 1](#) and [2](#) hold. Let $C_i(t)$ be continuous. Consider the measurements (2) for system (1), assuming that $v_i(t)$ is colored Gaussian noise of the form*

$$\dot{v}_i(t) = -\lambda v_i(t) + v'_i(t), \quad (25)$$

where $v'_i(t)$ is white Gaussian noise and $\lambda > 0$ is the bandwidth parameter. Choose the event threshold $\varepsilon_i(t)$ in (4) such that it has a strictly positive minimum value $\varepsilon_{\min}(T) > 0$ for any compact interval $[0, T]$. Then, the sequence $\{\tau_k^i\}_{k=0}^\infty$ does not exhibit Zeno behavior w.p.1.

Proof. The dynamical system (1) follows a stochastic differential equation driven by a Wiener process. In particular, $w(t)$ is formally understood as the generalized derivative $w(t) = dw'(t)/dt$ of a Wiener process $w'(t)$. It is well known that the corresponding state process $x(t)$ admits a modification with continuous sample paths on any compact interval, w.p.1 [32]. The colored noise process (25) is also driven by a Wiener process and admits the same modification. Then, since $y_i(t) = C_i(t)x(t) + v_i(t)$ is composed of products and sums of continuous terms, it follows that $y_i(t)$ admits a modification with continuous sample paths on any compact interval, w.p.1. In the following, we work on such a continuous sample path, and the conclusions hold w.p.1.

Fix an arbitrary $T > 0$ and let $\varepsilon_{\min}(T) = \inf_{t \in [0, T]} \varepsilon_i(t) > 0$. Since $y_i(t)$ is continuous on $[0, T]$, it is uniformly continuous on $[0, T]$. Therefore, for the value $\varepsilon_{\min}(T) > 0$, there exists a $\Delta(T) > 0$ such that, for all $s, t \in [0, T]$, if $|t - s| < \Delta(T)$, then $\|y_i(t) - y_i(s)\| < \varepsilon_{\min}(T)$.

Consider any event time $s = \tau_k^i \in [0, T]$. For any $t \in (\tau_k^i, \min\{\tau_k^i + \Delta(T), T\})$, we have

$$\|y_i(t) - y_i(\tau_k^i)\| < \varepsilon_{\min}(T) \leq \varepsilon_i(t),$$

so the triggering condition in (4) cannot be satisfied in that interval. Hence,

$$\tau_{k+1}^i - \tau_k^i \geq \Delta(T) > 0,$$

whenever $\tau_{k+1}^i \leq T$. This excludes an infinite accumulation of event times in $[0, T]$. Since $T > 0$ is arbitrary, the sequence $\{\tau_k^i\}_{k=0}^\infty$ does not exhibit Zeno behavior, w.p.1. $\square$

Remark 6. [Theorem 3](#) has been established under additional assumptions, namely that $v_i(t)$ is modeled as colored noise rather than white Gaussian noise. This guarantees sufficient regularity of the trajectories, preventing the occurrence of infinitely many events in finite time caused by abrupt variations, since white Gaussian noise does not admit continuous sample paths. The colored noise model (25) approximates white noise as λ increases. Moreover, colored noise naturally arises in digital implementations with sample and hold mechanisms, which impose bandwidth limitations on the noise process. Hence, assuming colored noise with unknown bandwidth remains consistent with practical scenarios.

Theorem 4. *Let [Assumptions 1](#), [2](#), [5](#) and [6](#) hold, and assume that there is no Zeno behavior in the sequences $\{\tau_k^i\}_{k=0}^\infty$, $\forall i \in \mathcal{V}$. Consider the distributed filter (23) and the event-triggering mechanism (22), choosing the event threshold $\delta_i(t)$ such that it has a strictly positive minimum value $\delta_{\min}(T) > 0$ for any compact interval $[0, T]$. Then, the sequences $\{\ell_k^i\}_{k=0}^\infty$ do not exhibit Zeno behavior.*

Proof. The estimate $\hat{x}_i(t)$ in (23b) depends on event-triggered measurements and neighbor estimates. Define the combined update sequence

$$\{\zeta_k^i\}_{k=0}^\infty = \{\tau_k^i\}_{k=0}^\infty \cup \{\ell_k^j\}_{k=0}^\infty, \quad \forall j \in \mathcal{N}_i,$$

which collects all instants where measurements or neighbor estimates are updated. By assumption, the measurement transmission times $\{\tau_k^i\}_{k=0}^\infty$ are not Zeno. Hence, Zeno behavior in $\{\ell_k^i\}_{k=0}^\infty$ could only originate locally at some estimator node. Fix an interval $(\zeta_k^i, \zeta_{k+1}^i)$ where no external updates occur. On this interval, measurements and neighbor estimates are constant. Therefore, (23b) reduces to an ordinary differential equation with locally bounded right-hand side. In particular, there exists $B \geq 0$ such that

$$\|\dot{\hat{x}}_i(t)\| \leq B, \quad \forall t \in (\zeta_k^i, \zeta_{k+1}^i).$$

If $B = 0$, then $\|\dot{\hat{x}}_i(t)\| = 0$ on $(\zeta_k^i, \zeta_{k+1}^i)$, so $\hat{x}_i(t)$ is constant. Hence, $\|\hat{x}_i(t) - \hat{x}_i(\ell_k^i)\| = 0 < \delta_i(t)$, and no event can be triggered during this interval. When $B > 0$, assume that two consecutive triggering instants $\ell_k^i < \ell_{k+1}^i$ lie inside $(\zeta_k^i, \zeta_{k+1}^i)$. Moreover,

$$\|\hat{x}_i(\ell_{k+1}^i) - \hat{x}_i(\ell_k^i)\| \leq \int_{\ell_k^i}^{\ell_{k+1}^i} \|\dot{\hat{x}}_i(s)\| ds \leq B(\ell_{k+1}^i - \ell_k^i).$$

From the triggering condition (22),

$$0 < \delta_{\min}(T) \leq \delta_i(\ell_{k+1}^i) \leq \|\hat{x}_i(\ell_{k+1}^i) - \hat{x}_i(\ell_k^i)\|$$

for any $T \geq \ell_{k+1}^i$. Combining both inequalities yields

$$\ell_{k+1}^i - \ell_k^i \geq \frac{\delta_{\min}(T)}{B} > 0.$$

Hence, a strictly positive lower bound on inter-event times holds inside $(\zeta_k^i, \zeta_{k+1}^i)$, excluding Zeno accumulation within such intervals. Finally, since solutions of (23b) are continuous, accumulation cannot occur at the update instants $\{\zeta_k^i\}_{k=0}^\infty$ either. Therefore, $\{\ell_k^i\}_{k=0}^\infty$ does not exhibit Zeno behavior. $\square$

7.2. Time regularization

A simple way to enforce a strictly positive minimum inter-event time is to introduce time regularization, namely, to disable the triggering condition for a fixed interval after each event. This approach is frequently adopted in stochastic settings, where rare noise realizations may otherwise induce arbitrarily fast successive events [33]. The triggering rules become

$$\begin{aligned} \tau_{k+1}^i &= \inf\{t \geq \tau_k^i + \underline{\tau} \mid \|\mathbf{y}_i(t) - \mathbf{y}_i(\tau_k^i)\| \geq \varepsilon_i(t)\}, \\ \ell_{k+1}^i &= \inf\{t \geq \ell_k^i + \underline{\ell} \mid \|\hat{x}_i(t) - \hat{x}_i(\ell_k^i)\| \geq \delta_i(t)\}, \end{aligned} \quad (26)$$

where $\underline{\tau} > 0$ and $\underline{\ell} > 0$ are regularization parameters. By construction,

$$\tau_{k+1}^i - \tau_k^i \geq \underline{\tau}, \quad \ell_{k+1}^i - \ell_k^i \geq \underline{\ell},$$

so a uniform minimum inter-event time is guaranteed.

From the perspective of exploiting implicit measurement information between events, disabling the triggering condition during the interval $(\tau_k, \tau_k + \underline{\tau})$ implies that such implicit information is not available over that short period. However, the purpose of exploiting implicit information is to prevent the growth of estimation uncertainty during long inter-event intervals without new measurements. Since $\underline{\tau}$ can be chosen arbitrarily small, neglecting implicit information over $(\tau_k, \tau_k + \underline{\tau})$ does not affect the intended behavior. Therefore, it is reasonable to treat implicit information as effectively available between events.

In digital implementations, continuous-time dynamics are approximated with a sampling step $h > 0$, and the triggering condition is evaluated once per step. Choosing $\underline{\tau} = h$ and $\underline{\ell} = h$ is therefore natural, since it does not impose additional constraints beyond those already induced by sampling. Under this choice, two situations may arise if the triggering condition is violated during the regularization interval:

- The violation disappears before the interval ends. No event is generated. This corresponds to a transient excursion of the signal that does not persist over one sampling step, and the loss of information is no more significant than from the sampling itself.
- The violation persists after the interval ends. An event is generated immediately at the next admissible checking instant. In the extreme case of persistent violation, events occur periodically with period h , which coincides with full communication at every sampling step.

Remark 7. The use of positive event thresholds and time regularization parameters to guarantee well posedness prevents exact convergence to the centralized filter, as established in item 2 of Theorems 1–2. Indeed, simultaneously setting the event thresholds and the time regularization parameters to zero may lead to Zeno behavior. Nevertheless, these parameters can be chosen arbitrarily small, which avoids Zeno behavior while yielding a performance close to that of the centralized filter. The limit case $\varepsilon_{\max} \rightarrow 0$, $\delta_{\max} \rightarrow 0$ is mainly of theoretical interest, as it characterizes convergence to the centralized solution. In practice, selecting zero thresholds is undesirable since it removes any communication savings.

8. Simulation experiments

We include a simulation experiment to illustrate the performance of the proposed filter. Without loss of generality, we focus here on an LTI system, since this is the case we have considered for the distributed filter (23) under multi-channel event-triggered communication. However, recall that the results for the filter (14) under event-triggered measurements hold for LTV systems as well.

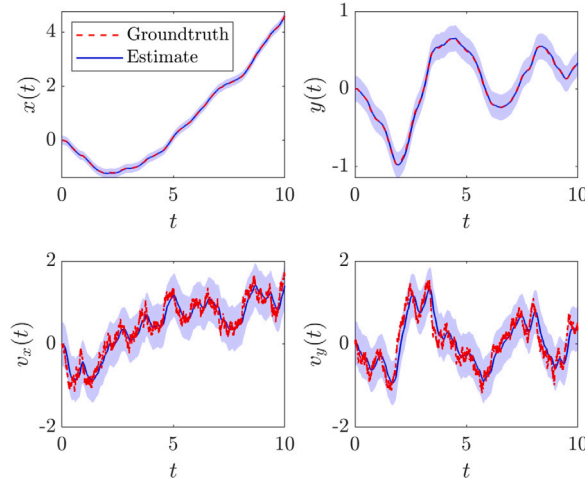


Fig. 2. State estimates with the baseline centralized Kalman-Bucy filter with full measurement data.

We consider a 2D integrator system, where the state $\mathbf{x}(t) = [x(t), y(t), v_x(t), v_y(t)]^\top$ represents the 2D Cartesian position coordinates $x(t)$, $y(t)$, and the velocity components $v_x(t)$, $v_y(t)$ in both axes. The system dynamics are described by (1) along with

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 \end{bmatrix}. \quad (27)$$

The plant is observed by a network of $N = 6$ estimator nodes, with the topology shown in Fig. 1, noting that the communication graph for the set of estimator nodes running the distributed filter is given by the dotted lines. As depicted in dashed lines in Fig. 1, each estimator node receives a sensor measurement from its corresponding i th sensor. The sensor models are given by (2) along with $\mathbf{C}_1 = \mathbf{C}_5 = [1, 0, 0, 0]$, $\mathbf{C}_2 = \mathbf{C}_4 = [0, 1, 0, 0]$, $\mathbf{C}_3 = [0, 0, 1, 0]$, $\mathbf{C}_6 = [0, 0, 0, 1]$, and $\mathbf{V}_i = 0.01$, $\forall i \in \mathcal{V}$. To simulate the system dynamics, we use the Euler–Maruyama discretization with step $h = 10^{-3}$.

Each estimator node runs the event-triggered filter (23), where we assume that the constant matrix $\mathbf{P}_\infty$ has been precomputed offline, as explained in Remark 2. Therefore, estimator nodes transmit only their state estimates $\hat{\mathbf{x}}_i(t)$ to their neighbors, and measurements are transmitted to the estimators, at a series of events. Due to the digital implementation, note that the discretization step h naturally acts as the time regularization parameters τ , ℓ in the event-triggering mechanisms (26) to avoid Zeno behavior. We set $\kappa = 5000$ for the distributed filter, recalling that $\kappa \rightarrow \infty$ improves its estimation performance. In the following experiments, we test the performance of the filter for different values of the event thresholds ε_i , δ_i , assuming for simplicity that they are constant and that the same value is used at all nodes.

We aim to verify that the distributed filter (23) under multi-channel event-triggered communication can achieve comparable performance to the centralized Kalman-Bucy filter with the complete measurement (3) of the network, which provides the optimal state estimate in this context, while achieving communication savings in both channels. As intermediate cases, we will also include the performance of the centralized filter 4 and of the distributed filter (14) under event-triggered measurements only (which corresponds to setting $\delta_i = 0$ in the simulation).

Figs. 2 and 3 show a qualitative comparison of the estimates obtained using the centralized filter and the distributed filter under multi-channel event-triggered communication with event thresholds $\varepsilon_i = 0.1$ and $\delta_i = 0.005$, for the same sequence of states and measurements. The distributed filter is initialized randomly to verify its convergence in the case of having unknown initial conditions, and the estimates $\hat{\mathbf{x}}_i(t)$ computed by all nodes are shown in Fig. 3. The blue band around the estimates represents twice the standard deviation, using the values of the estimated covariance matrix $\mathbf{P}_\infty$, in analogy to a confidence interval for the estimates. Note that the nodes in the distributed case exhibit a high level of agreement in their estimates, resembling the performance obtained for the centralized filter in Fig. 2. However, thanks to the introduction of event-triggered communication, the distributed filter achieves this result while using only 48% of the communication slots for the sensor channel and 37% in the estimator one, on average (considering that full usage of each channel means that every device transmits data at all simulation steps).

In addition, we provide a numerical comparison of the different setups to validate their performance. As performance metrics, we consider the mean squared error (MSE) for the simulation at each estimator, E , and the communication rates C_s and C_e in the sensor and estimator communication channels, respectively. We compute them as follows:

$$E = \frac{1}{NT} \sum_{i=1}^N \int_0^T \|\mathbf{x}(t) - \hat{\mathbf{x}}_i(t)\|^2 dt, \quad C = \frac{h}{NT} \sum_{i=1}^N e_i, \quad (28)$$

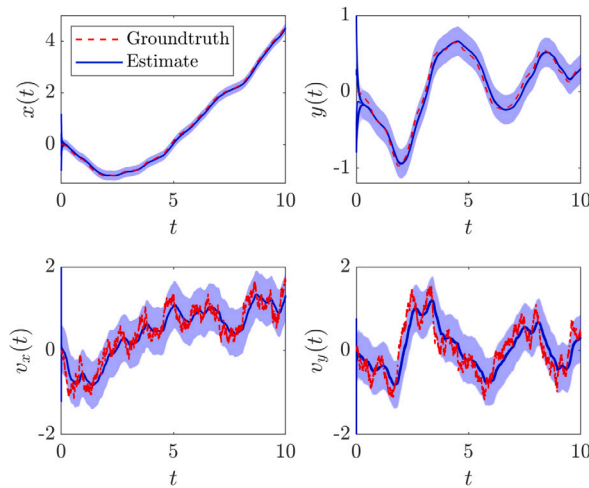


Fig. 3. State estimates obtained by each estimator node with our proposed distributed filter under multi-channel event-triggered communication, with $\kappa = 5000$, $\epsilon_i = 0.1$ and $\delta_i = 0.005$. The sensors use 48% of communication slots, while the estimators use 37%.

Table 1

Average results of 100 runs for centralized and distributed estimators under full and event-triggered communication.

		$\text{tr}(\mathbf{P}_\infty)$	E	C_s	C_e
Centralized Kalman-Bucy filter		0.1534	0.0716	1	–
Centralized filter, event-triggered measurements	$\epsilon_i = 0.1$	0.1789	0.0829	0.4817	–
	$\epsilon_i = 0.15$	0.2071	0.0953	0.2918	–
Our distributed filter ($\delta_i = 0$)	$\epsilon_i = 0$	0.1534	0.0794	1	1
	$\epsilon_i = 0.1$	0.1789	0.0897	0.4817	1
	$\epsilon_i = 0.15$	0.2071	0.1013	0.2918	1
Our distributed filter ($\epsilon_i = 0.1$)	$\delta_i = 0.005$	0.1789	0.1237	0.4817	0.4508
	$\delta_i = 0.01$	0.1789	0.2529	0.4817	0.2177

with T being the length of the experiment and e_i the number of events triggered by the i th sensor (for C_s) or estimator (for C_e). For the centralized case, $N = 1$ is used to compute E , since there is only one estimator. In the distributed case, note that E represents the average MSE for one estimator node within the network, recalling that the goal is for each node to obtain similar estimates to a centralized filter. Moreover, the communication rates are normalized between zero, representing no transmissions, and one, representing the case where each device transmits at all steps in the simulation. Due to the stochastic nature of the observed system and measurements, we perform 100 runs of length $T = 10$ for each method, such that the result is representative of the expected behavior for different noise realizations. Each method is run over the same 100 realizations of noise to ensure a fair comparison, and all filters are initialized to $\hat{\mathbf{x}}_i(0) = \mathbf{x}_0$.

The results of this experiment are shown in Table 1, considering different values of the event thresholds ϵ_i , δ_i in the event-triggered cases. We include the metrics E , C_s , C_e for each method, as well as $\text{tr}(\mathbf{P}_\infty)$, which represents the theoretical MSE estimated by the filter. Note that the case with $\epsilon_i = 0$, $\delta_i = 0$ matches the digital implementation of the full-communication distributed filter where each node transmits its data at every simulation step, so we include this limit case in the experiment. The simulations confirm our theoretical results. For our distributed filter, the case with $\epsilon_i = 0$, $\delta_i = 0$ recovers a comparable performance to the centralized Kalman-Bucy filter with full measurements. As the threshold ϵ_i is increased, the estimation error increases slightly, but significant communication savings are achieved in the sensor communication channel. Moreover, the distributed filter still obtains similar performance to the centralized one under the same event threshold ϵ_i . Then, by simultaneously increasing the threshold δ_i for the transmission of state estimates, the communication in the estimator channel can be similarly reduced, while still providing stable estimates with a bounded MSE. Thus, the results in Table 1 evidence a trade-off between accuracy and communication rate where, by tuning the event thresholds, the performance of the centralized filter can be approximated in a distributed fashion while reducing the transmission load in all channels of the network. The resulting values of E in the experiments are also smaller than the estimated value given by $\text{tr}(\mathbf{P}_\infty)$ for each ϵ_i in most cases, except when setting higher values of δ_i for the event-triggering mechanism in the estimators, so the estimated covariance $\mathbf{P}_\infty$ is a reasonable indicator of the expected quality of the estimates for a given threshold ϵ_i . Notice that E does not exactly match $\text{tr}(\mathbf{P}_\infty)$ even for the centralized Kalman-Bucy filter, which can be due to using a finite number of noise realizations to compute E , while the expected error $\text{tr}(\mathbf{P}_\infty)$ is obtained theoretically by considering the noise probability distributions.

9. Conclusions

We have presented a distributed filter for stochastic systems under event-triggered measurement sampling and communication between estimator nodes. To improve the estimates between sampling events and reduce uncertainty for large inter-event intervals, the estimators leverage implicit information in *not* receiving a new measurement, incorporating it as a virtual measurement. Through formal analysis and simulation experiments, we verify that our solution achieves stable estimates in the mean squared error sense and that undesirable Zeno behavior is avoided. Moreover, we have shown that, by tuning the consensus gain and the event thresholds, the performance of the distributed filter can be made arbitrarily close to that of the optimal centralized filter with full information of the network measurements, while significantly reducing the communication demands of the distributed setup.

CRedit authorship contribution statement

Irene Perez-Salesa: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Rodrigo Aldana-López:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Carlos Sagüés:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] S. He, H.-S. Shin, S. Xu, A. Tsourdos, Distributed estimation over a low-cost sensor network: A review of state-of-the-art, *Inf. Fusion* 54 (2020) 21–43.
- [2] S. Battilotti, F. Cacace, M. d'Angelo, A. Germani, Asymptotically optimal consensus-based distributed filtering of continuous-time linear systems, *Automatica* 122 (2020) 109189.
- [3] S. Battilotti, F. Cacace, M. d'Angelo, A stability with optimality analysis of consensus-based distributed filters for discrete-time linear systems, *Automatica* 129 (2021) 109589.
- [4] I. Perez-Salesa, R. Aldana-López, C. Sagüés, ODEFTC: Optimal distributed estimation based on fixed-time consensus, *Inf. Fusion* 116 (2025) 102783.
- [5] G. Battistelli, L. Chisci, D. Selvi, A distributed Kalman filter with event-triggered communication and guaranteed stability, *Automatica* 93 (2018) 75–82.
- [6] D. Selvi, G. Battistelli, Distributed Kalman filtering with adaptive communication, *IEEE Control. Syst. Lett.* 9 (2025) 15–20.
- [7] I. Perez-Salesa, R. Aldana-López, C. Sagüés, Event-triggered distributed state estimation based on asymptotic Kalman-Bucy filter, in: 2024 European Control Conference, ECC, 2024, pp. 731–736.
- [8] X. Ge, Q.-L. Han, X.-M. Zhang, L. Ding, F. Yang, Distributed event-triggered estimation over sensor networks: A survey, *IEEE Trans. Cybern.* 50 (3) (2020) 1306–1320.
- [9] Z. Kowalczyk, M. Domżański, Asynchronous distributed state estimation based on a continuous-time stochastic model, *Internat. J. Adapt. Control Signal Process.* 26 (5) (2012) 384–399.
- [10] Y. Hu, X. Lin, K. Peng, Distributed consensus filtering over sensor networks with asynchronous measurements, *Internat. J. Adapt. Control Signal Process.* 39 (1) (2025) 101–115.
- [11] Y. Hu, X. Lin, Weighted average consensus filtering for continuous-time linear systems with asynchronous sensor measurements, *IEEE Sensors J.* 25 (4) (2025) 6881–6893.
- [12] S. Wang, Y.-J. Pan, M. Guay, Distributed state estimation for linear time-invariant systems with aperiodic sampled measurement, *IEEE Trans. Control. Netw. Syst.* 11 (4) (2024) 1835–1844.
- [13] L. Yan, X. Zhang, Z. Zhang, Y. Yang, Distributed state estimation in sensor networks with event-triggered communication, *Nonlinear Dynam.* 76 (1) (2014) 169–181.
- [14] C. Zhang, Y. Jia, Distributed Kalman consensus filter with event-triggered communication: Formulation and stability analysis, *J. Franklin Inst.* 354 (13) (2017) 5486–5502.
- [15] N. Xia, F. Yang, Q.-L. Han, Distributed event-triggered networked set-membership filtering with partial information transmission, *IET Control Theory Appl.* 11 (2) (2017) 155–163.
- [16] X. Ge, Q.-L. Han, Distributed event-triggered H-infinity filtering over sensor networks with communication delays, *Inform. Sci.* 291 (2015) 128–142.
- [17] L. Ding, G. Guo, Distributed event-triggered H-infinity consensus filtering in sensor networks, *Signal Process.* 108 (2015) 365–375.
- [18] S. Trimpe, R. D'Andrea, Event-Based State Estimation With Variance-Based Triggering, *IEEE Trans. Autom. Control* 59 (12) (2014) 3266–3281.
- [19] S. Trimpe, Event-based state estimation: An emulation-based approach, *IET Control Theory Appl.* 11 (11) (2017) 1684–1693.
- [20] Q. Voortman, D. Efimov, A.Y. Pogromsky, J.P. Richard, H. Nijmeijer, An event-triggered observation scheme for systems with perturbations and data rate constraints, *Automatica* 145 (2022) 110512.
- [21] M. Miśkowicz, Event-based sampling strategies in networked control systems, in: 10th IEEE Workshop on Factory Communication Systems, 2014, pp. 1–10.
- [22] I. Perez-Salesa, R. Aldana-López, C. Sagüés, Bounding uncertainty in state estimation under dynamic event-triggered communication, *IEEE Trans. Syst. Man, Cybern.: Syst.* 55 (1) (2025) 209–220.
- [23] J. Sijs, B. Noack, U.D. Hanebeck, Event-based state estimation with negative information, in: 16th International Conference on Information Fusion, International Society of Information Fusion (ISIF), 2013, pp. 2192–2199.
- [24] G. Battistelli, A. Benavoli, L. Chisci, Data-driven communication for state estimation with sensor networks, *Automatica* 48 (5) (2012) 926–935.
- [25] E.J. Schmitt, B. Noack, W. Krippner, U.D. Hanebeck, Gaussianity-preserving event-based state estimation with an FIR-based stochastic trigger, *IEEE Control. Syst. Lett.* 3 (3) (2019) 769–774.

- [26] E.J. Schmitt, B. Noack, Event-based multisensor fusion with correlated estimates, in: 2024 27th International Conference on Information Fusion, FUSION, 2024, pp. 1–8.
- [27] R.E. Kalman, R.S. Bucy, New results in linear filtering and prediction theory, *J. Fluids Eng. Trans. the ASME* 83 (1961) 95–108.
- [28] I. Perez-Salesa, R. Aldana-López, C. Sagüés, A note on optimal distributed state estimation for linear time-varying systems, 2025, arXiv:2510.18712.
- [29] F. Arscott, A. Filippov, Differential equations with discontinuous righthand sides: Control systems, in: *Mathematics and its Applications*, Springer Netherlands, 1988.
- [30] I. Shames, T. Charalambous, C.N. Hadjicostis, M. Johansson, Distributed network size estimation and average degree estimation and control in networks isomorphic to directed graphs, in: 2012 50th Annual Allerton Conference on Communication, Control, and Computing (Allerton), 2012, pp. 1885–1892.
- [31] R. Aldana-López, R. Aragüés, C. Sagüés, Quasi-exact dynamic average consensus under asynchronous communication and symmetric delays, *ISA Trans.* (2025).
- [32] K. Åström, Introduction to stochastic control theory, in: *Mathematics in science and engineering*, Academic Press, 1970.
- [33] X. Ge, Q.L. Han, X.M. Zhang, D. Ding, Dynamic event-triggered control and estimation: A survey, *Int. J. Autom. Comput.* 18 (6) (2021) 857–886.